Check Point 1

In this document, we describe our ER diagram and the reasoning behind it.

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ER Digram

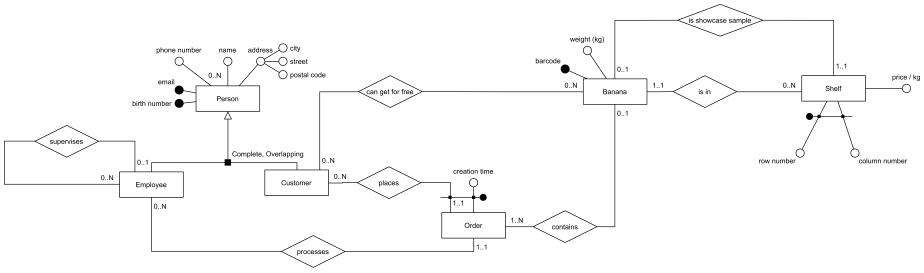

Figure 1: ER Diagram

Description

In this section we provide the description of entities and their relations.

Banana Central to our model is the **banana** entity. It has two attributes: a **barcode** (which identifies it) and a **weight (kg)** which is going to become useful once it is purchased. A banana *may* be **contained** inside *one* (we avoid multiple **customers** purchasing the same **banana**) **order**. *Every* banana is located in *one* (given its macroscopic properties, it may not be in two places at once) **shelf**. A **banana** may be **showcase sample** for *precisely one* **shelf**.

Shelf A shelf is identified with its **row** and **column** number inside the shop. It also has an attribute **price/kg** which is going to play a role in determining the prices of bananas in it. *Each* shelf has *one* (there is no need for multiple of them) **showcase sample banana**.

Order An order's only attribute is its **creation time**. It is a weak entity, identified by the **creation time** and the **customer** that **placed** it. It **contains** (at least *one*; an empty order is nonsensical) the **bananas** that were purchased. **Each** order has to be processed by an **employee** (letting multiple **employees process** *one* **order** would be ineffective).

Person A **person** is the base for the **customer** and **employee** entities. A **person** is identified by an **e-mail address** *or* a **birth number**. This entity also has a **name**, **address** (composed of **city**, **street**, and **postal code**), and *zero or more* **phone numbers**. *Each* person must be an **employee** or a **customer** (there is no good reason to include other **persons** in our model). Some **persons** may be both **employees** and **customers** (this allows for the punchline where an **employee** starts eating all the **bananas**).

Employee An **employee** *may* (we consider the owner to be an employee and they do not have any **supervisor**) have a **supervisor** who is also an **employee**.

Customer A **customer** *may* **place** *any number* (they become a potential **customer** once they enter the shop; not when they buy something) of **orders**. Some **customers** have "special rights" and they **can get** *some* (there is no reason to limit their special rights to just one banana) **bananas for free**. There also may exist multiple customers that have the "special right" that they **can get** *one specific banana* **for free** which is why the relation is *N:M*.